

**Teaching Unit «Molecular mechanisms of cell signaling »
M1-S1
2025**

Nuclear Receptors signaling

martine.glorian@sorbonne-universite.fr
regine.hepp@sorbonne-universite.fr

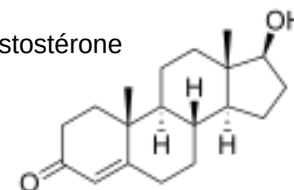
Nuclear receptors (NR) : definition

Intracellular receptors that function as transcription factors with an activity regulated by specific hydrophobic ligands.

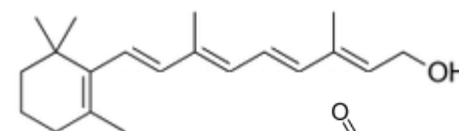
Common nuclear receptors and their ligands

Receptor	Abbreviation	Ligand
Androgen receptor	AR	Testosterone
Estrogen receptor	ER	Estrogen
Estrogen-related receptor	ERR	?
Glucocorticoid receptor	GR	Cortisol
Mineralocorticoid receptor	MR	Aldosterone
Progesterone receptor	PR	Progesterone
Retinoic acid receptor	RAR	Retinoic acid
Retinoid orphan receptor	ROR	?
Retinoic acid-related receptor	RXR	Rexinoids
Liver X receptor	LXR	Oxysterols
Peroxisome proliferator-activated receptor γ	PPAR γ	Fatty acid metabolites
Thyroid hormone receptor	TR	Thyroid hormone
Vitamin D ₃ receptor	VDR	Vitamin D ₃

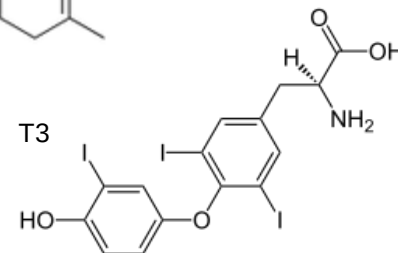
testostérone



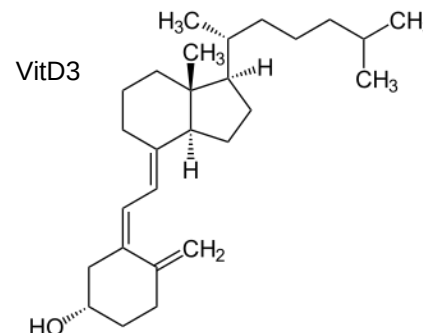
retinol



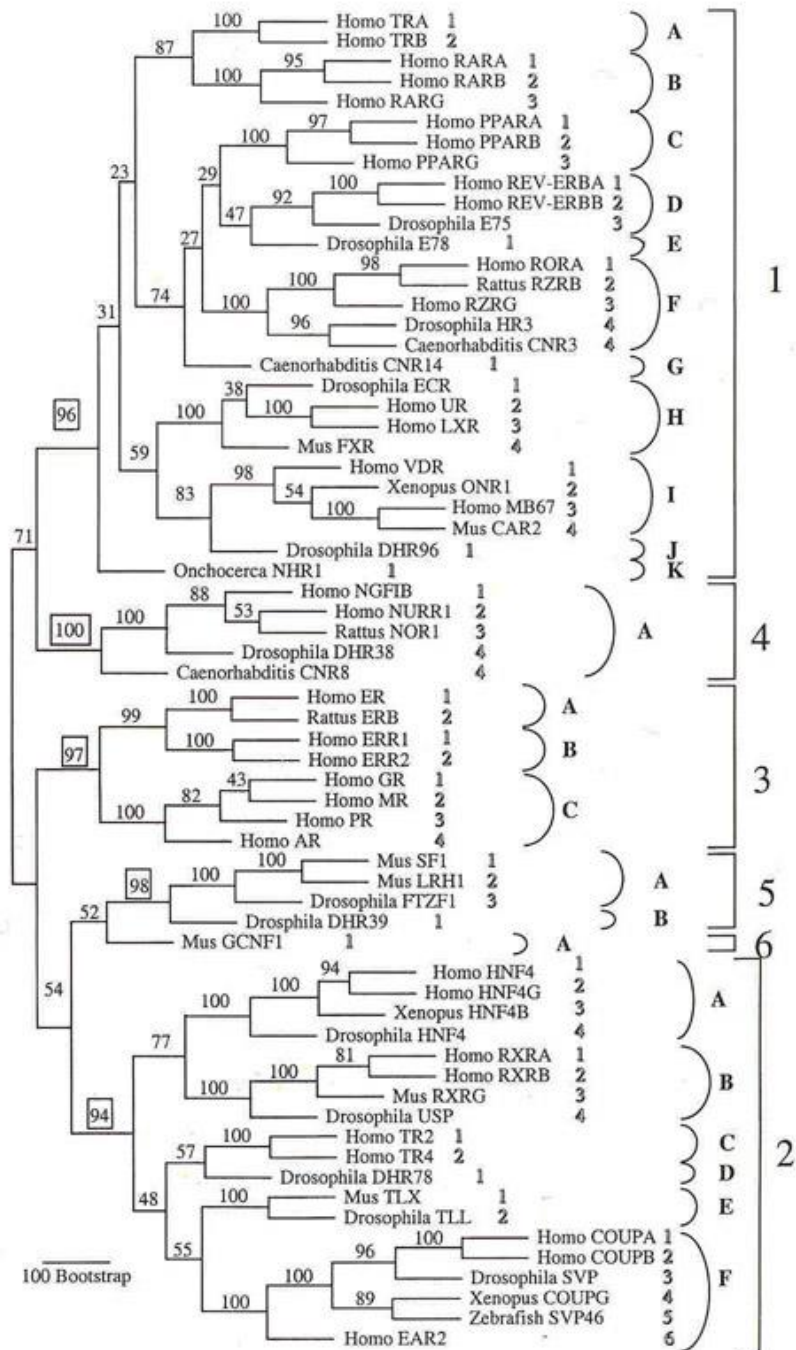
T3



VitD3



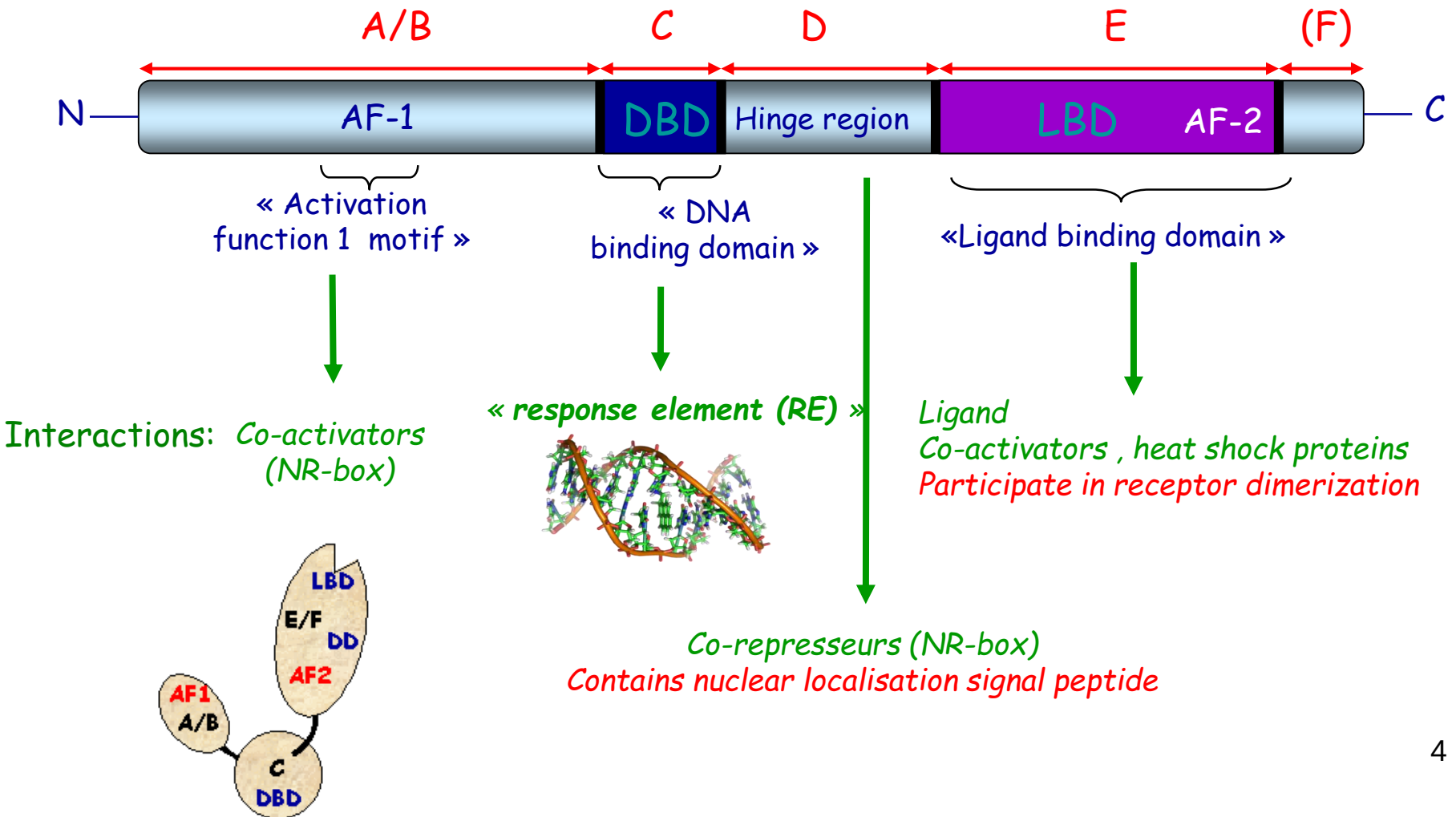
Classification by sequence homology



Androgen receptor	AR	NR3C4
Thyroid hormone receptor α	TR α	NR1A1

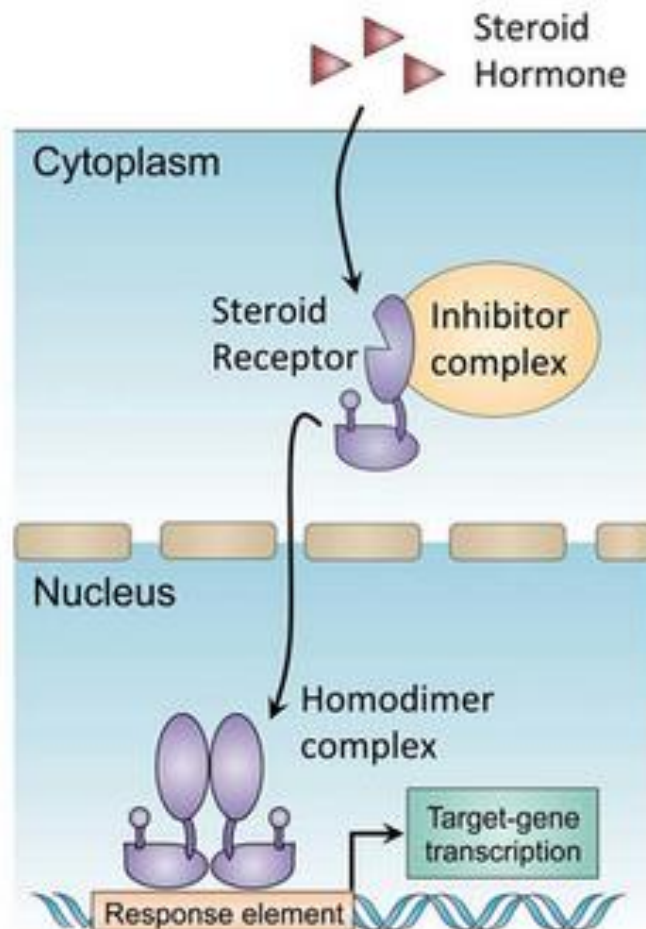
Functional organization

As transcription factors, NR bind DNA, ligands and regulate the recruitment of the transcription machinery

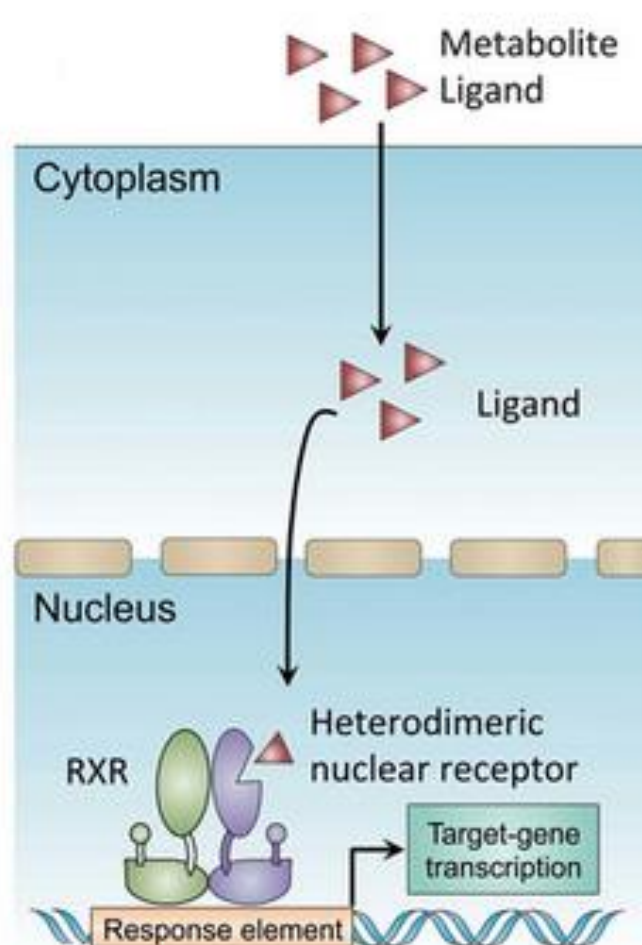


How do NR regulate transcription? 2 types of NR

TRANSLOCATION



NO TRANSLOCATION

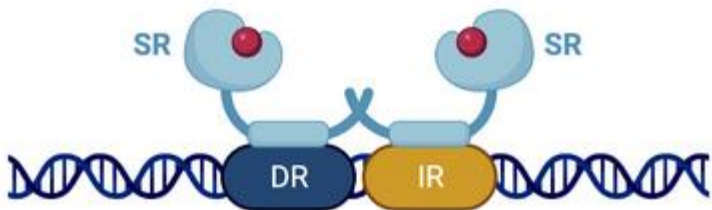


Thyroid hormones
Vitamines A, D
Fatty Acids,

How do NR regulate transcription? Receptor binding to DNA

Class I: Steroid Receptors (SR)

- Glucocorticoid
- Mineralocorticoid
- Progesterone
- Androgen
- Estrogen

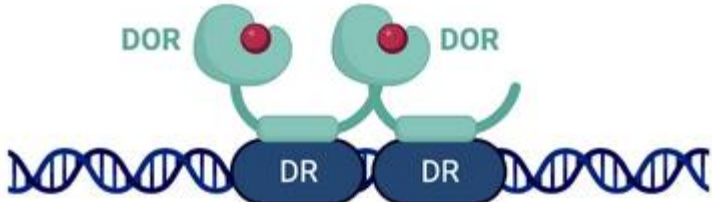


DR: direct repeat; IR: inverted repeat

palindrome
AGAACCA nnnTGGTTCT

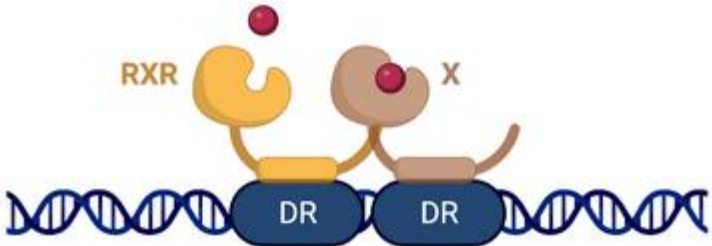
Class III: Dimeric Orphan Receptors (DOR)

- RXR (usp)
- HNF-4
- COUP/ARP



Class II: Retinoid X Receptor (RXR) Heterodimers

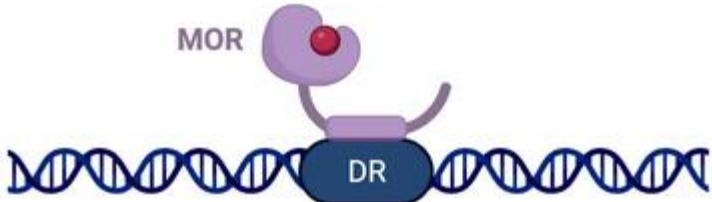
- Thyroid hormone
- PPAR_{α,β,γ}
- RAR_{α,β,γ}
- VDR
- EcR



AGGTCA n AGGTCA

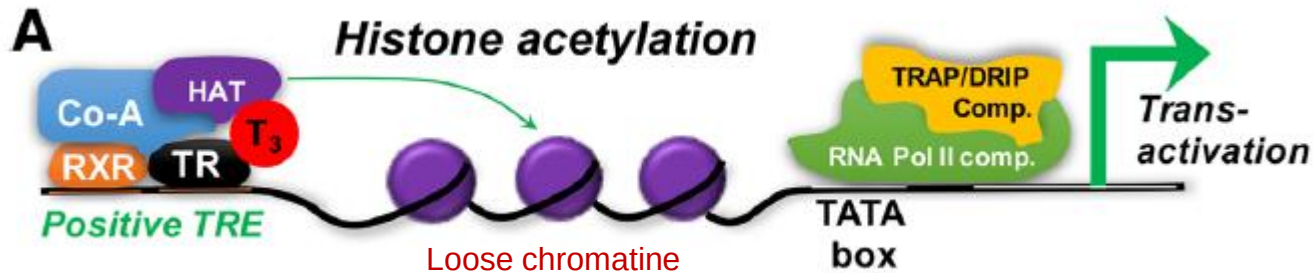
Class IV: Monomeric Orphan Receptors (MOR)

- NGFI-B (CEB-1')
- ELP/SF-1 (Ftz-F1)

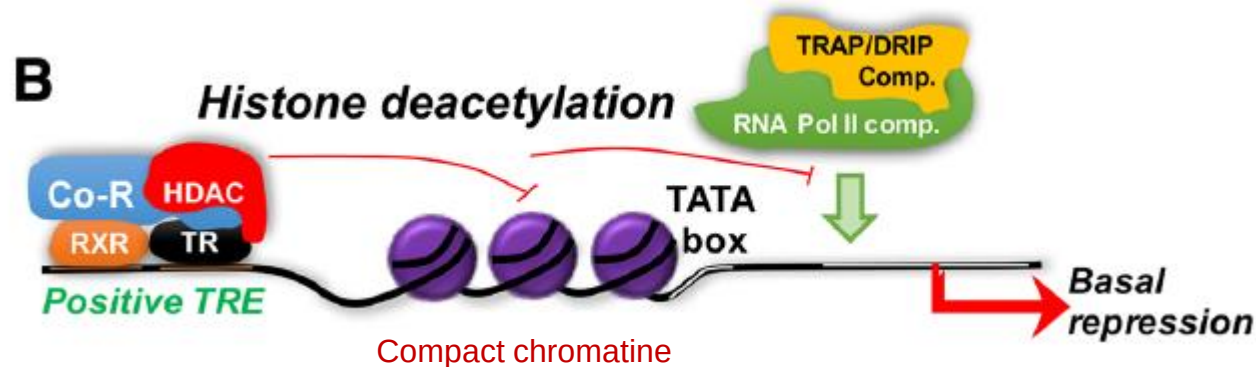


How do NR regulate transcription? Chromatin compaction

1. Activation in presence of ligand

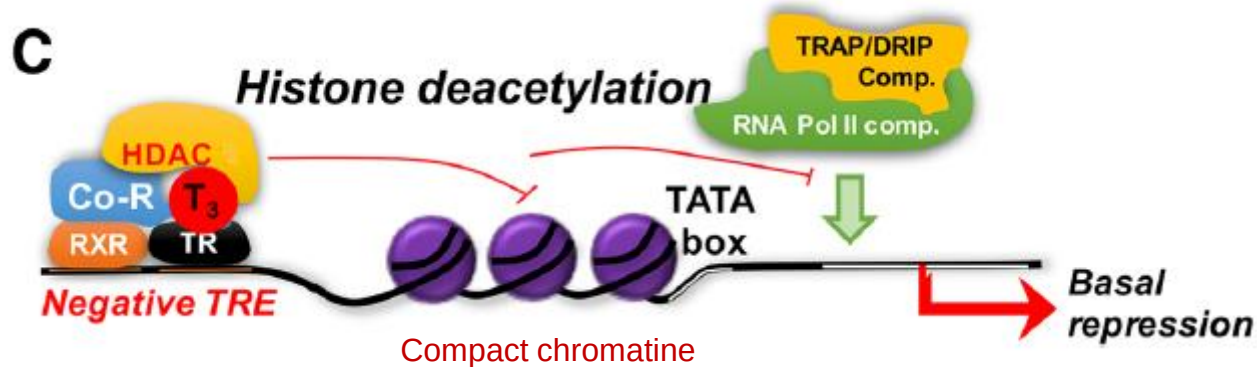


In **presence** of ligand, NR is bound to **co-activator proteins (Co-A)** that recruit histone acetyl transferase (HAT). Acetylation of the histones, opens the chromatine and allows the RNA polymerase to bind the TATA box and initiate the transcription of the target gene.



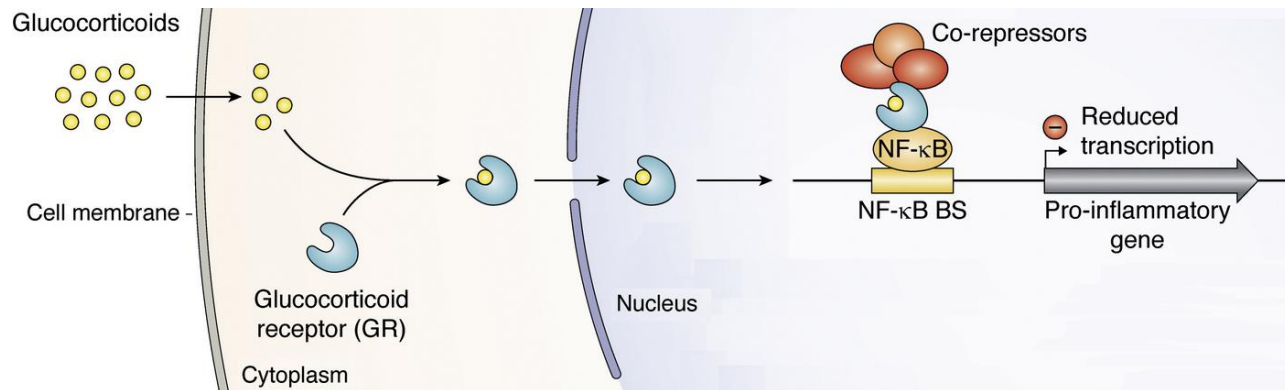
In absence of ligand, NR is bound to **co-repressor proteins (Co-R)** that recruit histone deacetylase (HDAC). Deacetylation of the histones, compacts the chromatine and inhibits the RNA polymerase to bind the TATA box and thus inhibits the transcription of the target gene.

2. Inhibition in presence of ligand



For genes repressed in presence of ligand, NR binds to **co-repressor proteins** that recruit histone acetyl deacetylase (HDAC). Histones deacetylation compacts the chromatin and inhibits binding of RNA polymerase to the TATA box. Transcription of the gene is inhibited.

3. Interaction with other transcription factors



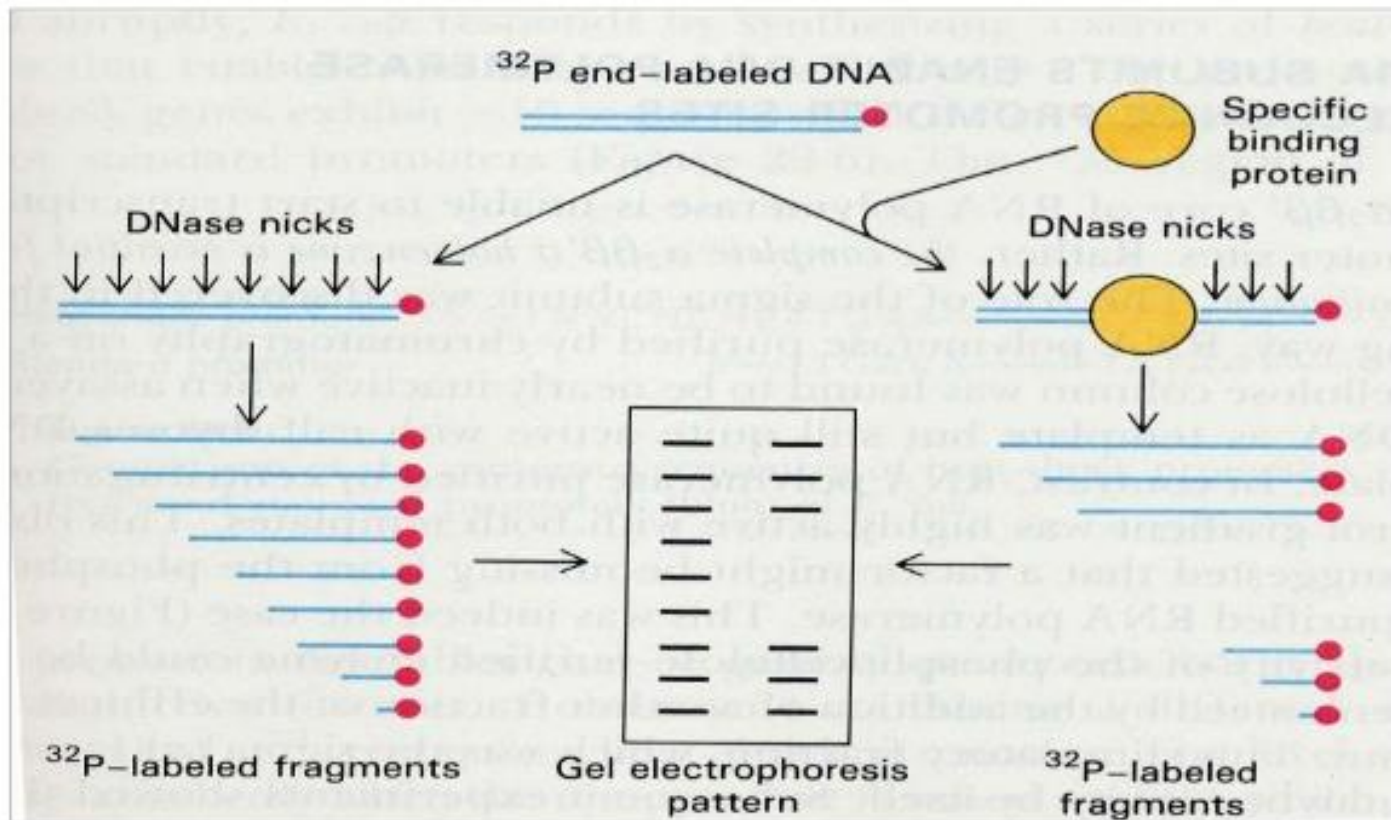
NR can interact with other transcription factors to regulate their transcription activity. For example Glucocorticoids inhibit the expression of inflammatory genes mediated by NF-κB through binding of GR and co-repressors to NF-κB

Methods for studying NR

- Protein interactions (see TDM3: protein interactions pull downs, yeast two hybrid....)
- Protein DNA interaction: chromatine IP (ChIP), gel shift assay, foot printing
- Transcription : use of a reporter gene controled by a promoter containing the HRE, RT-qPCR, RNA-seq

DNA footprinting: the promoter sequence is amplified by PCR, labeled, and incubated with nuclear proteins that bind to the promoter (if any), thereby protecting the DNA from the action of DNase. The fragments are deposited on gels. The white bands (footprints) mark the parts of the sequence that have bound proteins.

Nuclease protection foot printing

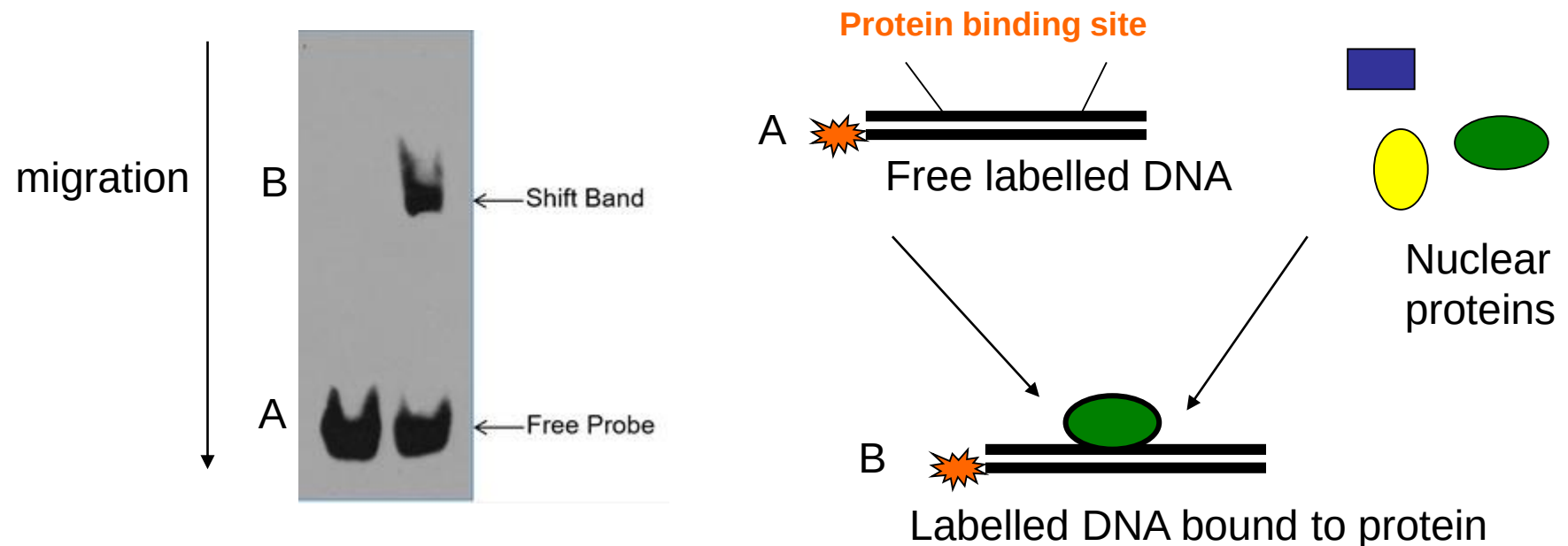


Gel shift assay

This method allows the detection of proteins or protein complexes binding to a given DNA sequence in the promoter region.

Nuclear proteins extracted from tissue of interest are incubated with a labelled DNA probes carrying the promoter region of interest.

Free DNA and DNA/protein complexed are separated on non-denaturing polyacrylamide gel



DNA-protein complexes are delayed (shifted) compared to the free probe

Analyzing a figure

Item
What models are used?
What are the experimental conditions?
What is being measured?
(Optional) How are the measurements made? or How is the experiment carried out?
What is observed?
What interpretation can be made from these observations?
So, what question does this figure answer?
What new question does this raise?

BACKGROUND

A transcriptional co-repressor that interacts with nuclear hormone receptors

J. Don Chen & Ronald M. Evans

Howard Hughes Medical Institute, Gene Expression Laboratory,
The Salk Institute for Biological Studies,
10010 North Torrey Pines Road, La Jolla, California 92037, USA

LETTERS TO NATURE

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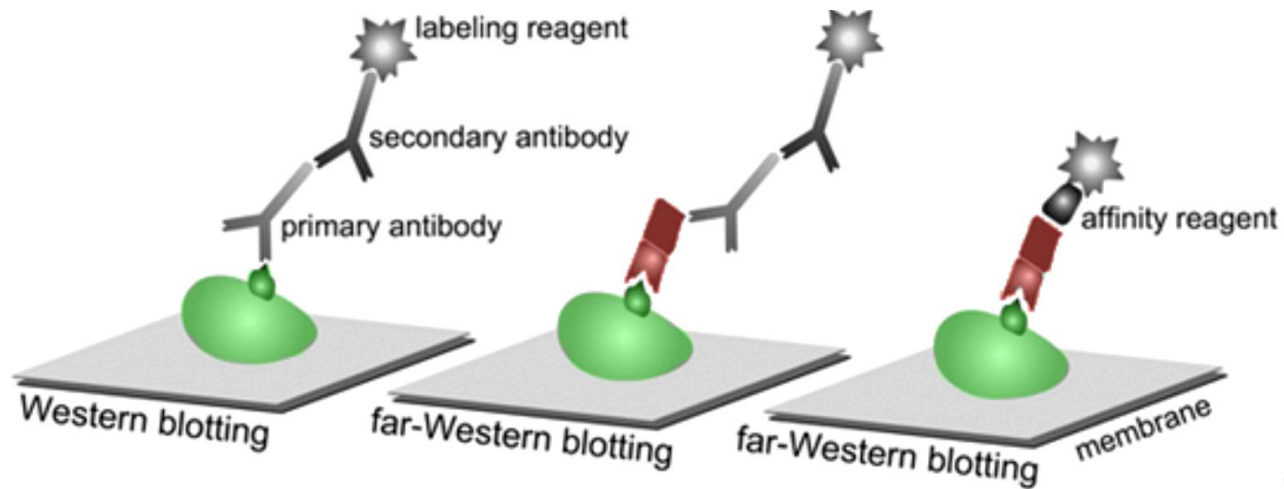
This article was published in 1995. At that time, researchers knew that transcriptional silencing by nuclear receptor exists and is important for many physiological processes. It was known that RAR and TR repress transcription in absence of ligand.

However, the molecular mechanism involved was unknown.

The authors searched for RN interacting proteins and performed experiments to decipher the implication of one of them, SMRT, in the regulation of transcription.

FAR WESTERN BLOTTING

Far-western blotting, is a method based on the technique of western blot to detect protein-protein interaction *in vitro*. **Far-western blot uses a non-antibody probe** which can bind the protein of interest. Thus, whereas western blotting is used for the detection of certain proteins, **far-western blotting is employed to detect protein/protein interactions**.



Protein of interest

Interacting protein (or domain) labelled or not

Groupe 1

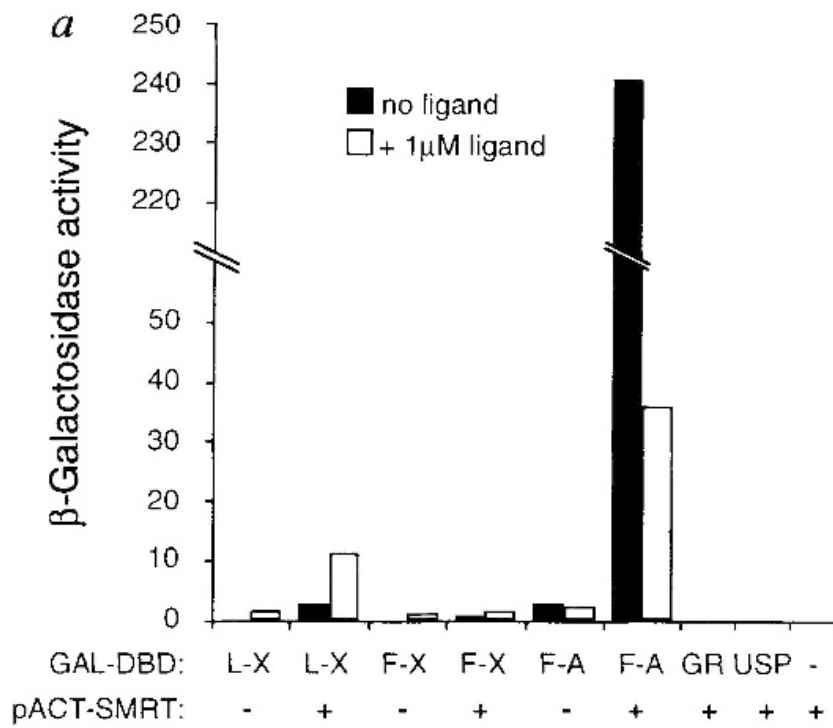
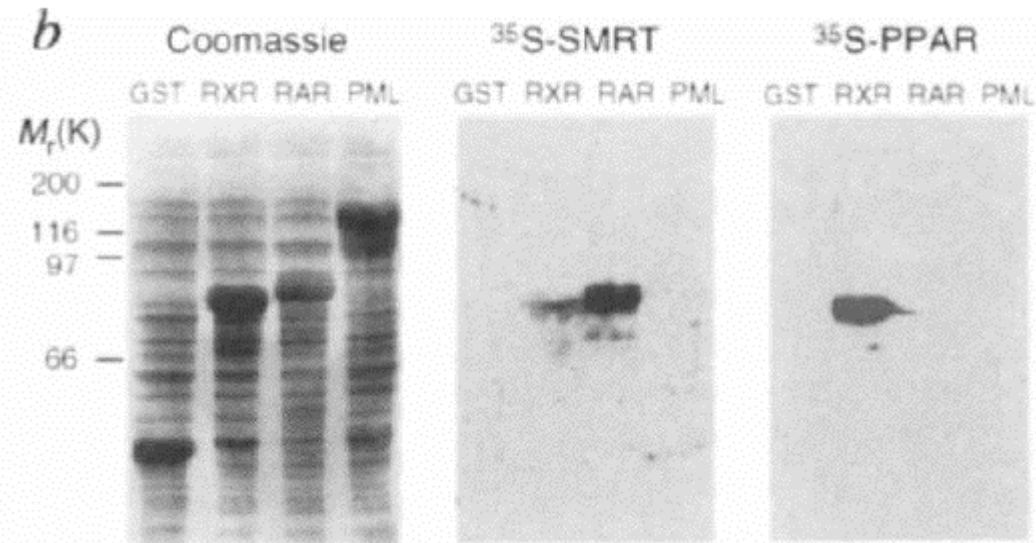
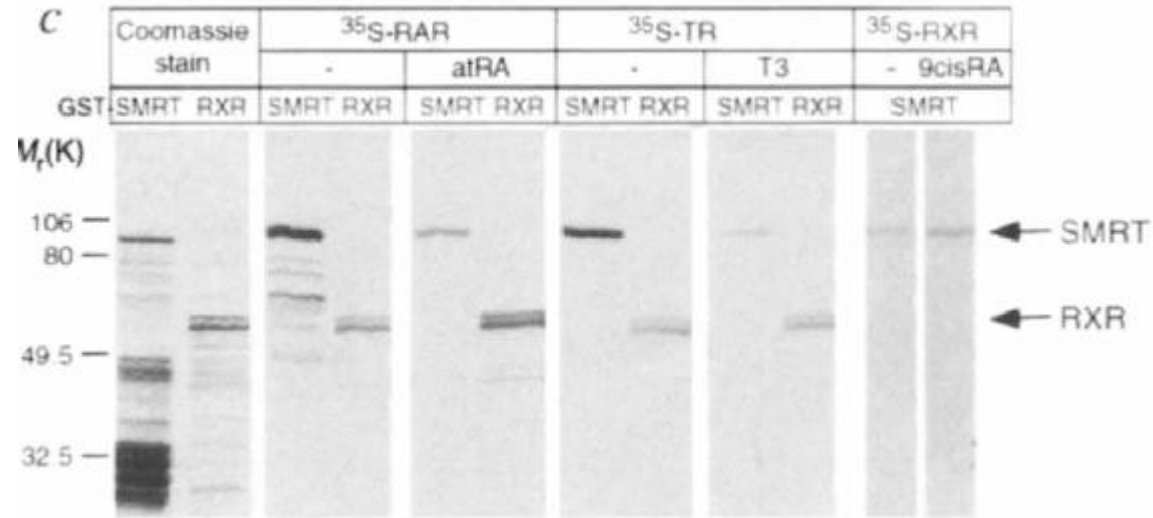


FIG. 1 a, Interaction of SMRT with receptors in yeast two-hybrid system. β -Galactosidase activities were quantified by liquid assay; data represent an average of three independent transformants. L, LBD; F, full-length; X, human RXR α ; A, human RAR α . b, Specific interaction of SMRT with GST-RAR and RXR in a far-western assay. As a control, mouse PPAR α complexes with GST-RXR but not with GST-RAR or control extracts.



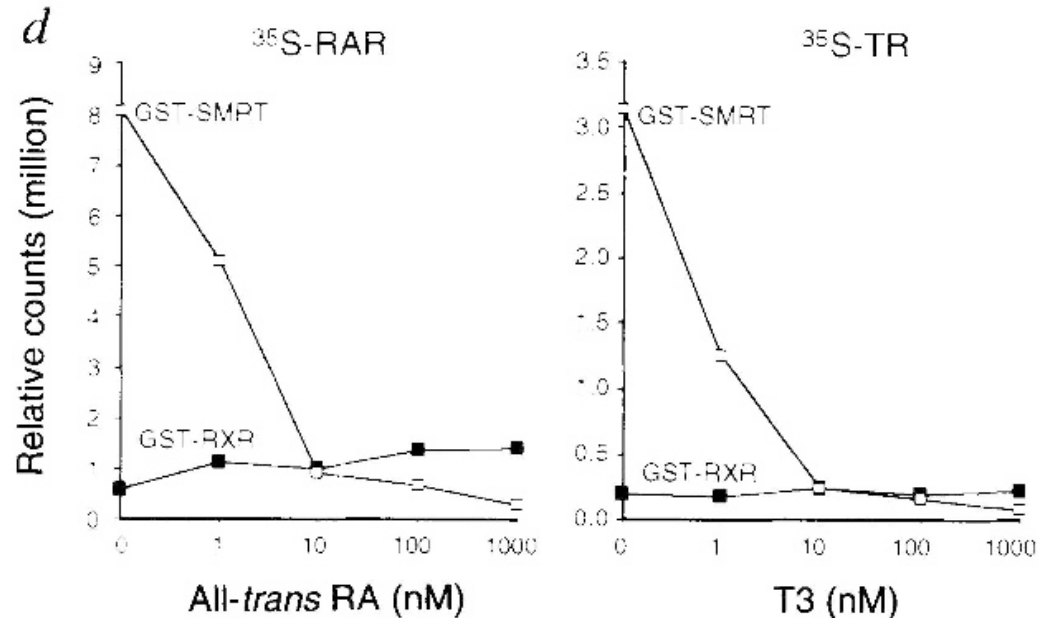
METHODS. Far-western blot. Total bacteria extracts expressing GST fusions of hRAR α (residues 156–462) or hRXR α LBD (residues 228–462) and control extracts expressing GST alone or GST-PML fusion protein were subjected to SDS-PAGE and electroblotted onto nitrocellulose and processed for far-western analysis as described³⁰. *In vitro* translated ³⁵S-labelled proteins were diluted into hybridization buffer and the filters were hybridized overnight at 4 °C without or with 1 μM of indicated ligand. After three washes with hybridization buffer, filters were dried and exposed for autoradiography or quantified by phosphor-imager. GST-SMRT is a GST fusion of the C-SMRT encoded by the yeast two-hybrid clone and, although it has been purified, it contains several degradation products. For yeast two-hybrid screening, a construct expressing the GAL4 DBD-hRXR α LBD (amino acids 198–462) fusion protein was used to screen a human lymphocyte cDNA library as described⁴. The full-length SMRT cDNA was isolated from a human HeLa cDNA library using the two-hybrid insert as a probe.

Groupe 2



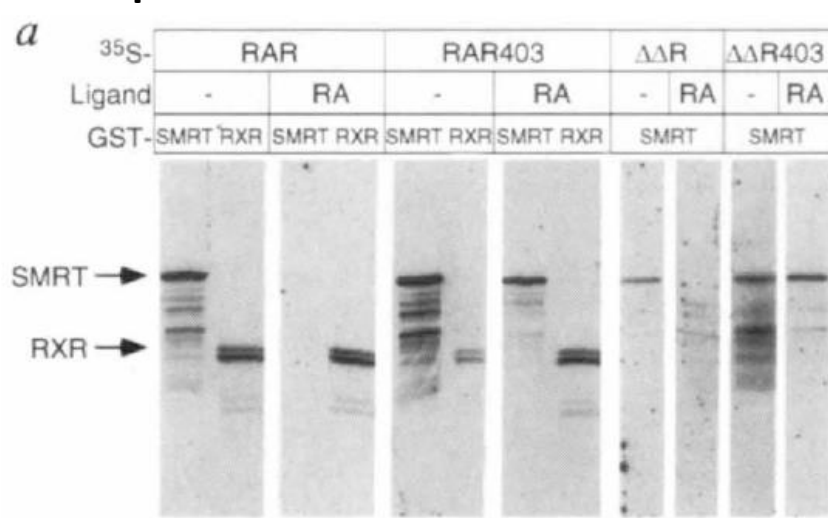
Authors have identified SMRT as an interacting protein of the RAR and TR receptors. They use far western blotting to study this interaction.

c, Interaction of full-length receptors with GST-SMRT in a far-western assay. d, Quantification of the ligand-dependent dissociation of RAR and TR from GST-SMRT by phosphorimager.

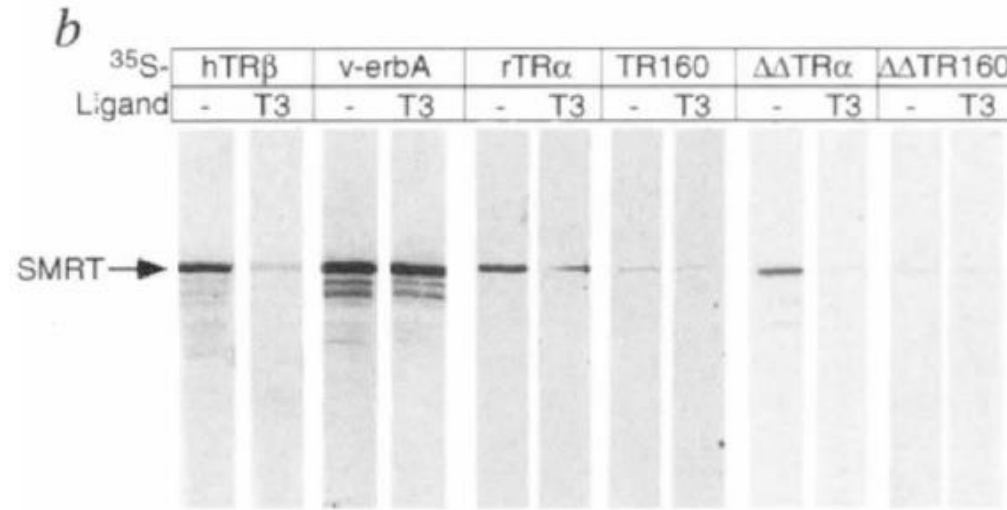


METHODS. Far-western blot. Total bacteria extracts expressing GST fusions of hRAR α (residues 156–462) or hRXR α LBD (residues 228–462) and control extracts expressing GST alone or GST-PML fusion protein were subjected to SDS-PAGE and electroblotted onto nitrocellulose and processed for far-western analysis as described³⁰. *In vitro* translated ³⁵S-labelled proteins were diluted into hybridization buffer and the filters were hybridized overnight at 4 °C without or with 1 μ M of indicated ligand. After three washes with hybridization buffer, filters were dried and exposed for autoradiography or quantified by phosphorimager. GST-SMRT is a GST fusion of the C-SMRT encoded by the yeast two-hybrid clone and, although it has been purified, it contains several degradation products.

Groupe 3



a) RAR403, Ct deletion mutant, dominant negative with strong repression; ΔΔR, Nt deletion mutant ΔΔR 403 double mutant



b) v-erbA, a dominant negative thyroid hormone receptor. No activation upon ligand binding. TR160: no inhibitory property in absence of ligand but can stimulate transcription in presence of ligand

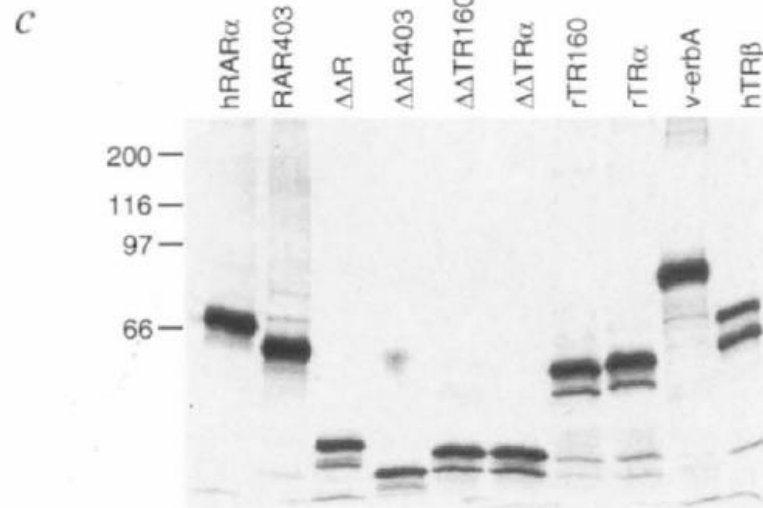


FIG. 2 a. Interaction of wild-type hRARα and deletion mutant RAR403 with SMRT. ³⁵S-labelled receptors were used as probes to hybridize immobilized GST-SMRT in the presence (10 μM) or absence of all-trans retinoic acid. The total bacteria extract expressing GST-RXR was included as a control. When quantified by phosphorimager, RAR403 shows a 4-fold stronger interaction with SMRT than does wild-type RAR. b. Interaction of the oncogenic v-ErbA and rTRα R160 mutant (TR160) with GST-SMRT in a far-western assay. When quantified by phosphorimager, the v-ErbA shows an 18-fold stronger interaction with SMRT than hTRβ, and the TR160 mutant shows a 10-fold lower signal than the rTRα. c. Autoradiograph showing the ³⁵S-labelled receptors on SDS-PAGE. RAR403: hRARα amino acids (aa) 1–403; ΔΔR: hRARα aa 154–462; ΔΔR403: hRARα aa 157–403; ΔΔTR160: LBD of rTRα with Pro 160 to Arg mutation; ΔΔTRα: rTRα aa 122–410; TR160: full-length rTRα with Pro 160 to Arg mutation.

Authors have shown that SMRT interacts with RAR and TR. They use far western blotting of receptors variants to study this interaction.

Groupe 4

The authors have shown that SMRT binds RAR and TR in absence of ligand by using yeast two hybrid and far western blotting. They have also shown that SMRT is not bound to these receptors in presence of the ligand.

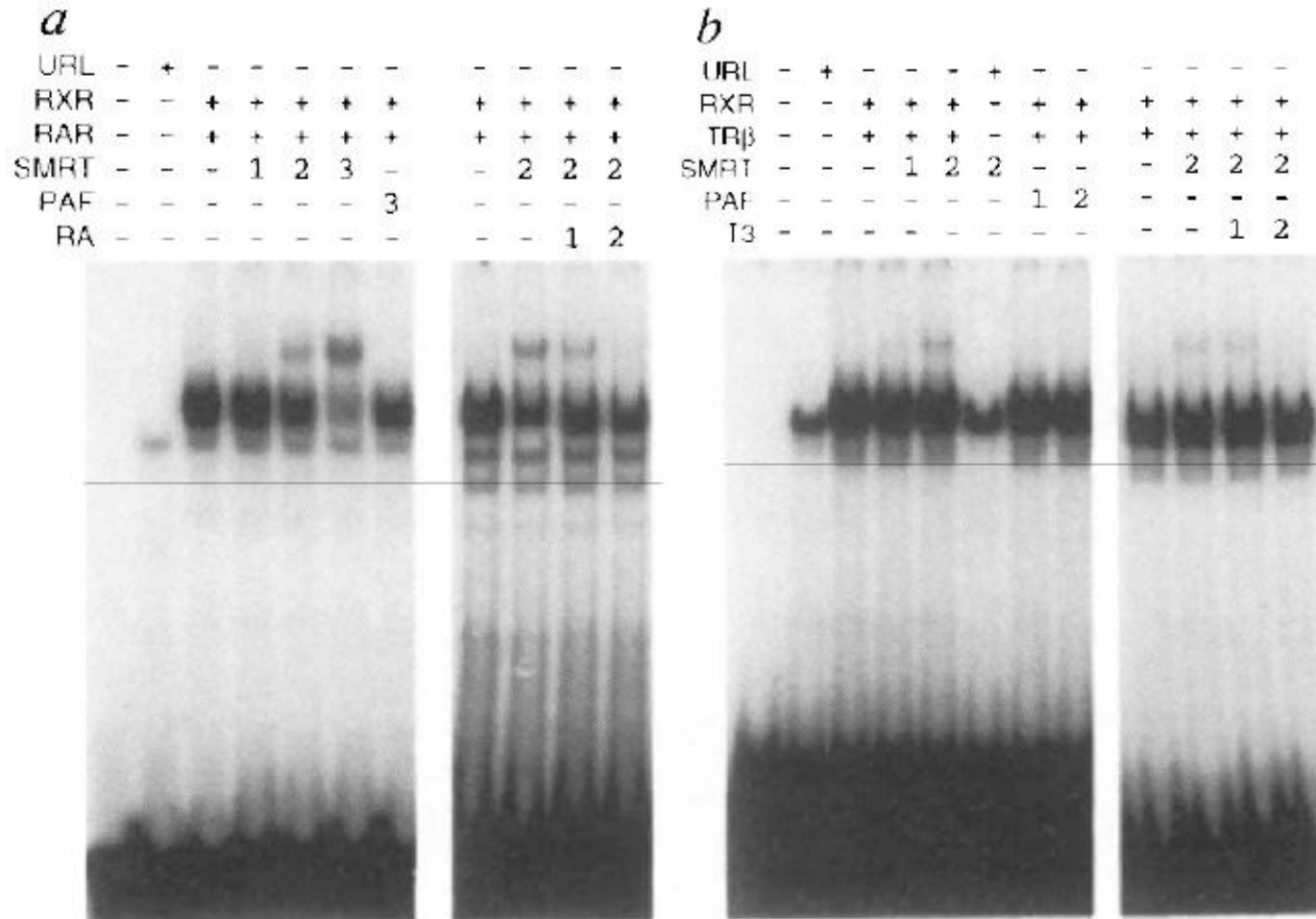


FIG. 3 Interaction of SMRT with receptor-DNA complex and the effect of ligands. **a**, Interaction of SMRT with RXR-RAR heterodimer on a DR5 element comprising AGGTCA direct repeat spaced by five nucleotides in a gel retardation assay. The amounts of GST fusion protein added in this experiment are (1), 6 ng; (2), 60 ng; and (3), 600 ng. PAF is a PML-associated protein isolated from another yeast two-hybrid screen (J.D.C. *et al.*, unpublished result). **b**, Interaction of SMRT with RXR-TR heterocomplex on a MLV TRE element. The amount of GST fusion protein added in this experiment are (1), 10 ng; (2) 100 ng. **METHODS.** Gel retardation assay: *in vitro* translated receptor or unprogrammed reticulocyte lysate (URL) was incubated with 1 μ g poly(dIdC) on ice for 15 min in a total of 20 μ l containing 75 mM KCl, 7.5% glycerol, 20 mM HEPES (pH 7.5), 2 mM DTT and 0.1% NP-40, with or without ligand as indicated (1, 10 nM; 2, 100 nM). A 32 P-labelled double-stranded oligonucleotide probe was added into the binding reaction (10,000 c.p.m. per reaction), and the reaction was further incubated for 20 min at room temperature. The protein-DNA complex was separated on a 5% native polyacrylamide gel at 150 volts.

Groupe 5 The authors have shown in vitro that SMRT binds RAR and TR in absence of ligand by using yeast two hybrid and far western blotting of purified proteins. They have also shown that SMRT is not bound to these receptors in presence of the ligand.

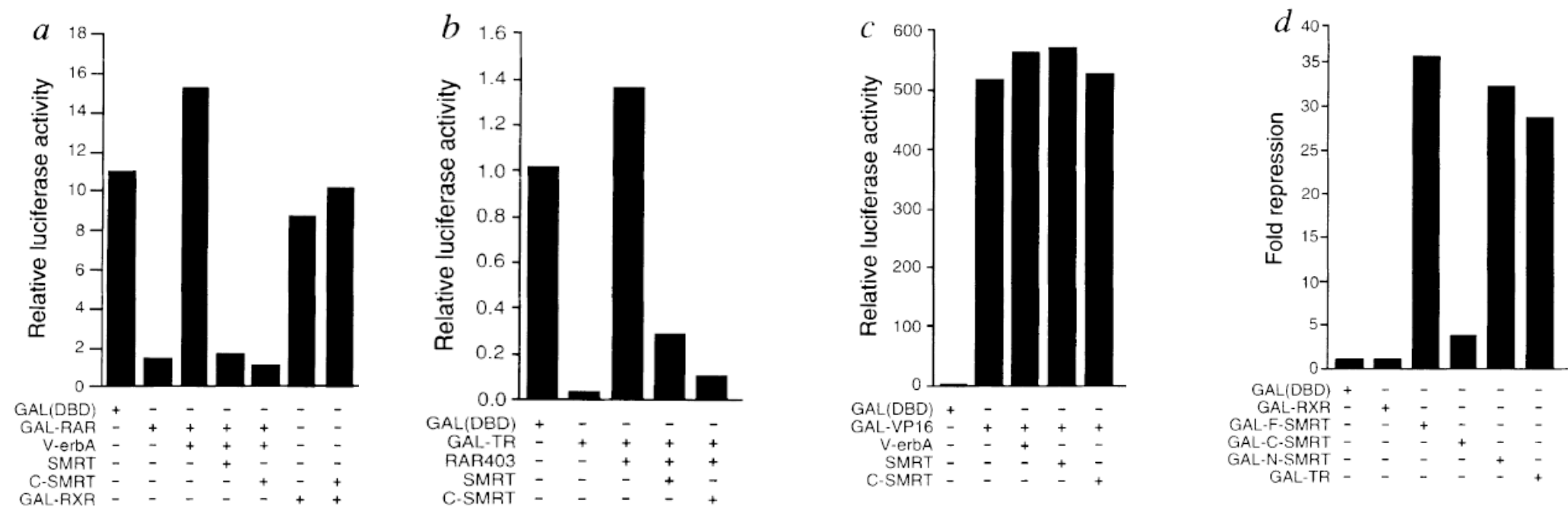


FIG. 4 Mediation of the silencing effect of hRAR α and hTR β by SMRT *in vivo*. *a*, Effect of v-ErbA and SMRT on the basal level promoter activity of GAL–RAR (GAL4 DBD–hRAR α 156–462) and GAL–RXR (GAL4 DBD fusion of hRXRa LBD). *b*, Effect of RAR403 truncation mutant and SMRT on the basal-level promoter activity of GAL–TR (GAL4 DBD–hTR β 173–456). *c*, v-ErbA and full-length SMRT or C-SMRT have no effect on GAL-VP16 activity. *d*, The intrinsic *trans*-repression domain of SMRT analysed by GAL4 DBD fusions. Fold repression was calculated by dividing the normalized luciferase activity transfected with the GAL4 DBD alone by those transfected with 30 ng GAL DBD fusion constructs. GAL-F-SMRT, GAL4 DBD fusion of full-length SMRT; GAL-N-SMRT, GAL4 DBD fusion of N terminus of SMRT (aa 1–981); GAL-C-SMRT, GAL4 DBD fusion of C-SMRT; GAL–TR, GAL4 DBD fusion of hTR β LBD.

METHODS. Transient transfection assay. The CV-1 cells were plated in 24-well plates at a density of 50,000 cells per well. The expression plasmids were transfected into cells by lipofection. In each transfection, 5 ng GAL–RAR and 15 ng v-erbA or SMRT were used together with 150 ng reporter construct containing 4 copies of GAL4 binding sites in front of a minimal thymidine-kinase promoter and a CMX- β -gal construct as an internal control. The relative luciferase activity was calculated by normalizing to the β -gal activity.

v-erbA, a thyroid hormone receptor with very strong inhibition of transcription

Groupe 6

Using, Yeast two hybrid, far western blotting and reporter gene assays, the authors have shown that SMRT binds RAR and TR receptor and acts as a co-repressor of the transcription in vitro in absence of receptor ligands.

One year later, the authors published a new article and confirmed their results in mammalian cells.

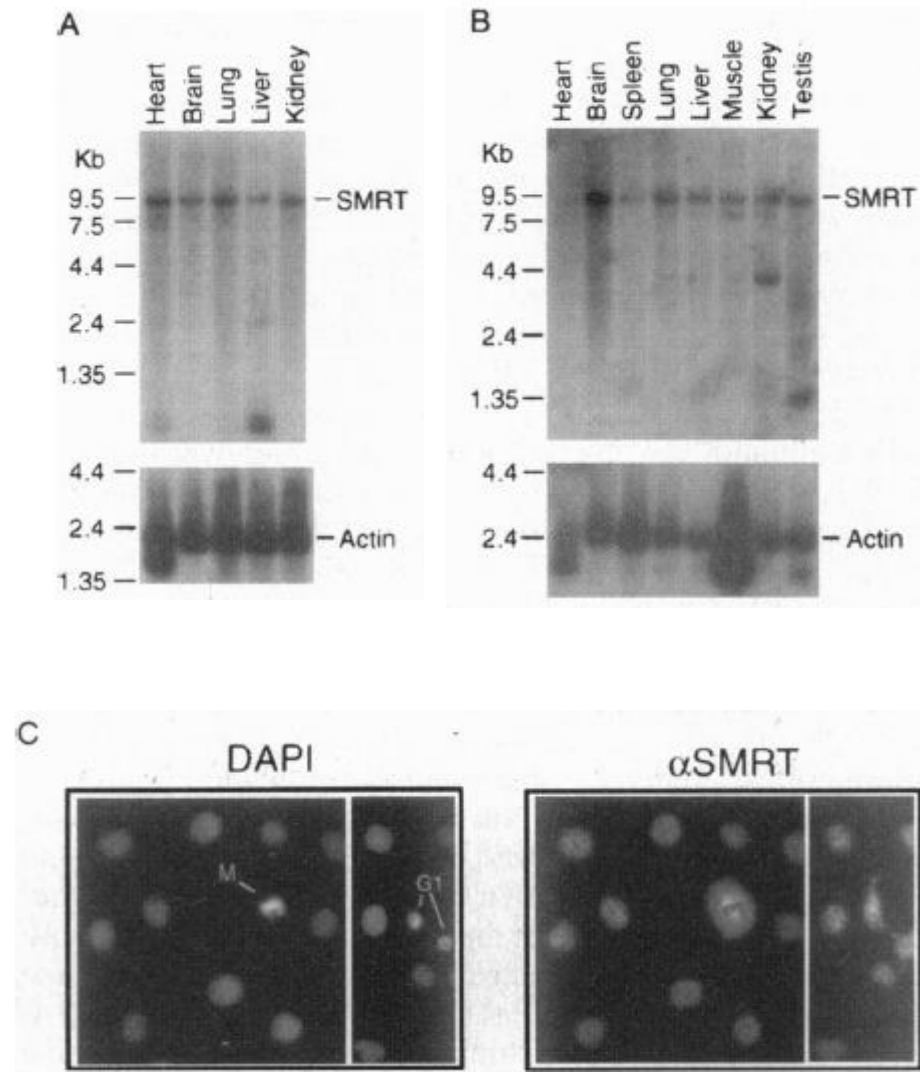


FIG. 4. SMRT is a ubiquitous nuclear protein. The expression pattern of SMRT in a human fetal Northern blot (A) and a mouse multiple tissue Northern blot (B) were analyzed as described in the text. The filters were stripped and rehybridized with a human actin probe (Lower). Note the existence of several smaller messages in addition to the full-length SMRT. (C) CV-1 cells grown on coverslips were processed for immunofluorescence analysis using a rabbit anti-SMRT polyclonal antibody (α SMRT). The same cells were counterstained by DNA binding dye DAPI for the location of the nucleus. One mitotic cell (M) and two cells in early G1 phase (G1) are indicated.

inated by changing the DNA binding specificities of the transfected receptors to those of the glucocorticoid receptor⁶. These altered-specificity RARs were then tested on promoters containing DR + 5 in which the downstream half-site was mutated to the glucocorticoid-receptor core recognition motif, AGAACA. The altered-specificity RARs containing CoR box mutations (L187P, A194P) were more active in response to 10^{-8} M all-*trans*-RA than the corresponding RARs containing a wild-type hinge region, indicating that interaction with N-CoR was not required for transactivation (Fig. 3c). Remarkably, mutant RARs lacking the ability to interact with N-CoR were capable

of activating transcription from a DR + 1-containing promoter, but the wild-type RAR was not (Fig. 3d). Together, these data indicate that the molecular explanation for repression by RAR on a DR + 1 element reflects the failure of ligand to cause the dissociation of the co-repressor, N-CoR, rather than altering co-activator association (Fig. 3e). These observations imply that interactions with N-CoR are dominant with respect to ligand-dependent transactivation, and show how positive or negative transcriptional effects of RAR/RXR heterodimers can be allosterically controlled by the DNA sequences to which they bind. □

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A transcriptional co-repressor that interacts with nuclear hormone receptors

J. Don Chen & Ronald M. Evans

Howard Hughes Medical Institute, Gene Expression Laboratory,
The Salk Institute for Biological Studies,
10010 North Torrey Pines Road, La Jolla, California 92037, USA

TRANSCRIPTIONAL silencing mediated by nuclear receptors is important in development, differentiation and oncogenesis^{1–3}. The mechanism underlying this effect is unknown but is one key to understanding the molecular basis of hormone action. Here we identify a receptor-interacting factor, SMRT, as a silencing mediator (co-repressor) for retinoid and thyroid-hormone receptors. SMRT is a previously undiscovered protein whose association with receptors both in solution and bound to DNA-response elements is destabilized by ligand. The interaction with mutant receptors correlates with their transcriptional silencing activities. *In vivo*, SMRT functions as a potent co-repressor, and a GAL4 DNA-binding domain fusion of SMRT behaves as a frank repressor of a GAL4-dependent reporter. Together, our results identify a new class of cofactors which may be important mediators of hormone action.

Using a GAL4 DNA-binding domain (DBD)–RXR fusion in a yeast two-hybrid screening⁴, we isolated several complementary DNA clones. One of these clones, SMRT, interacts strongly with unliganded full-length retinoic-acid receptor (RAR) but weakly or not at all with other receptors (Fig. 1a), and was selected for further characterization. In yeast, the ligand-binding domain (LBD) of the RAR and thyroid-hormone receptor (TR) is efficient in the interaction, whereas in both cases this interaction is reversed by ligand (data not shown). In a far-western analysis, SMRT preferentially complexes with immobilized RAR, marginally associates with the retinoid-X receptor (RXR) and shows no association with control polypeptides (Fig. 1b). In contrast, the peroxisomal proliferator-activated receptor (PPAR) selectively associates with RXR but not with RAR. Similarly, RAR or TR interacts strongly with immobilized

SMRT and RXR but not with glutathione-S-transferase (GST) or bacterial proteins, whereas RXR interacts weakly with SMRT (Fig. 1c). Binding of ligand to RAR or TR reduces their interactions with SMRT but not with RXR, and binding of ligand to RXR has slight effect (Fig. 1a, c). Figure 1d quantitates the dose-dependent release of SMRT from RAR or TR by ligand, which is very close to the dissociation constants^{5,6}.

Full-length SMRT encodes a polypeptide of 1,495 amino acids with an estimated relative molecular mass of 168,000 and is rich in proline and serine residues (Fig. 1e). GenBank comparison reveals similarity of the carboxy-terminal domain to a partial cDNA encoding another receptor-interacting protein, RIP13 (ref. 7) (Fig. 1e), whose role in receptor signalling is unknown. Northern blot analysis indicates that SMRT is ubiquitously expressed in low quantities and immunofluorescence data suggest that SMRT is a nuclear protein (data not shown).

Unlike other nuclear receptors, unliganded RAR and TR possess a strong silencing domain which represses basal-level promoter activity of their target genes^{8,9}. The preferential interaction of SMRT with RAR and TR in the absence of hormone suggests that SMRT may play a role in mediating the transcriptional silencing effect of receptors. To investigate further the involvement of SMRT in silencing, we tested its interactions with mutants that display distinct silencing and/or transactivation activities. Figure 2a shows that full-length and the LBD of RAR both associate with SMRT and that this association is blocked by ligand. The C-terminal-deletion mutant RAR403 is a potent suppressor of basal promoter activity of RAR target genes^{10,11}. As expected RAR403 and its amino-terminal-deletion derivative, Δ RAR403, interact strongly with SMRT in either the presence or absence of ligand, consistent with SMRT mediating the repressor activity of this mutant.

We next analysed the interactions of SMRT with two different classes of TR mutants: first, the oncogene *v-erbA*, which has strong silencing ability but no transactivation activity^{5,12,13}, and second, the Pro160→Arg point mutant *TR160*, which loses its capacity in basal-level suppression but retains hormone-dependent transactivation^{14,15}. Figure 2b shows that SMRT interacts strongly with *v-ErbA* both in presence or absence of ligand, but there is little interaction with *TR160* mutant. These data strongly support the idea that SMRT mediates transcriptional silencing

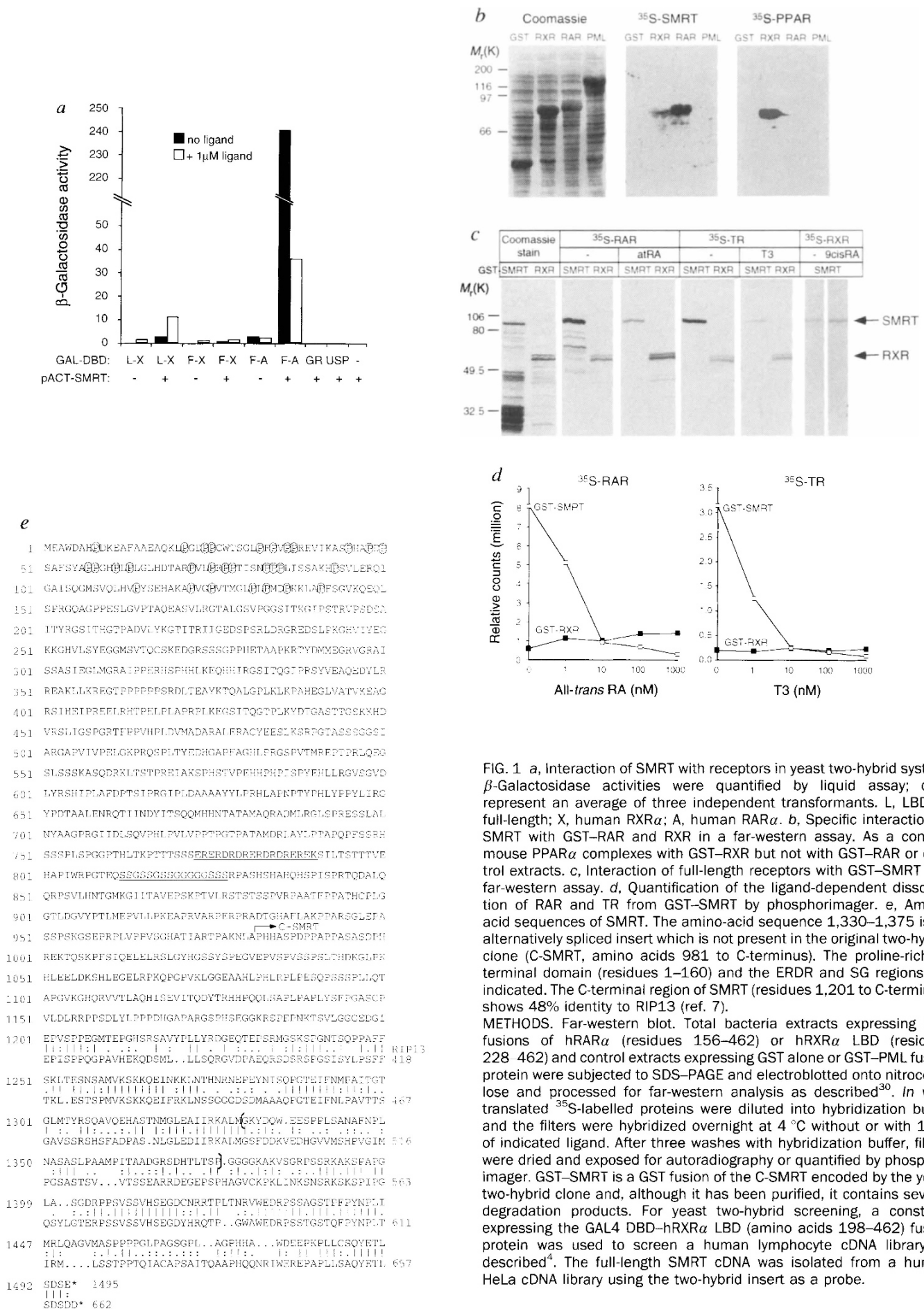
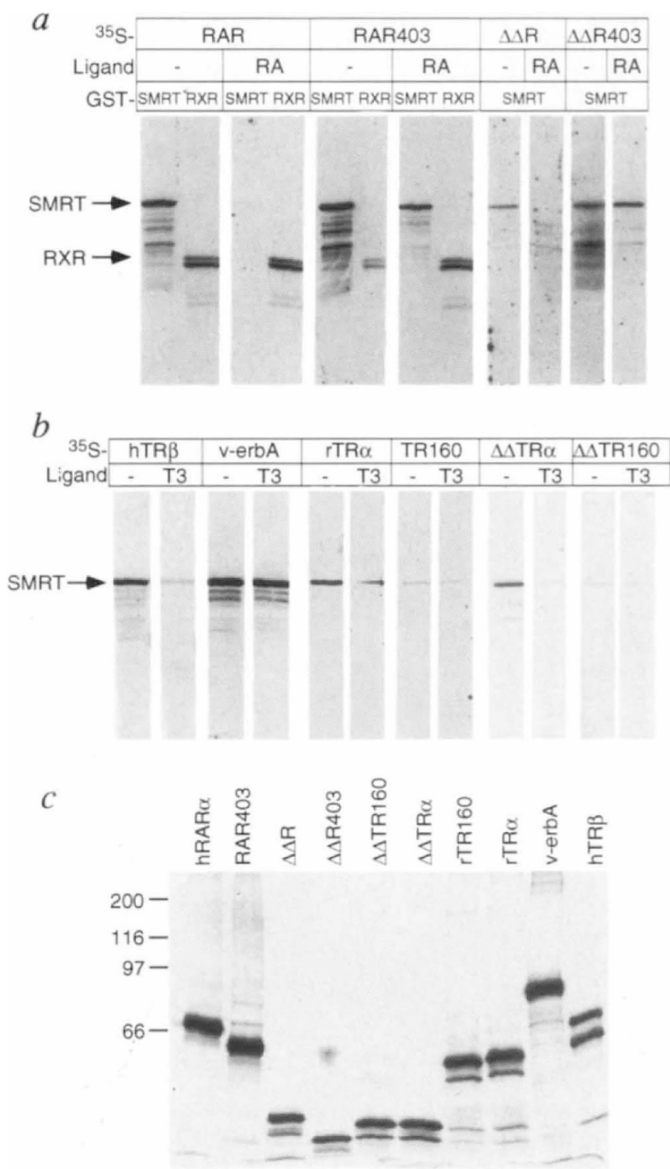


FIG. 1 *a*, Interaction of SMRT with receptors in yeast two-hybrid system. β -Galactosidase activities were quantified by liquid assay; data represent an average of three independent transformants. L, LBD; F, full-length; X, human RXR α ; A, human RAR α . *b*, Specific interaction of SMRT with GST-RAR and RXR in a far-western assay. As a control, mouse PPAR α complexes with GST-RXR but not with GST-RAR or control extracts. *c*, Interaction of full-length receptors with GST-SMRT in a far-western assay. *d*, Quantification of the ligand-dependent dissociation of RAR and TR from GST-SMRT by phosphorimager. *e*, Amino-acid sequences of SMRT. The amino-acid sequence 1,330–1,375 is an alternatively spliced insert which is not present in the original two-hybrid clone (C-SMRT, amino acids 981 to C-terminus). The proline-rich N-terminal domain (residues 1–160) and the ERDR and SG regions are indicated. The C-terminal region of SMRT (residues 1,201 to C-terminus) shows 48% identity to RIP13 (ref. 7).

METHODS. Far-western blot. Total bacteria extracts expressing GST fusions of hRAR α (residues 156–462) or hRXR α LBD (residues 228–462) and control extracts expressing GST alone or GST-PML fusion protein were subjected to SDS-PAGE and electroblotted onto nitrocellulose and processed for far-western analysis as described³⁰. *In vitro* translated ³⁵S-labelled proteins were diluted into hybridization buffer and the filters were hybridized overnight at 4 °C without or with 1 μ M of indicated ligand. After three washes with hybridization buffer, filters were dried and exposed for autoradiography or quantified by phosphorimager. GST-SMRT is a GST fusion of the C-SMRT encoded by the yeast two-hybrid clone and, although it has been purified, it contains several degradation products. For yeast two-hybrid screening, a construct expressing the GAL4 DBD-hRXR α LBD (amino acids 198–462) fusion protein was used to screen a human lymphocyte cDNA library as described⁴. The full-length SMRT cDNA was isolated from a human HeLa cDNA library using the two-hybrid insert as a probe.



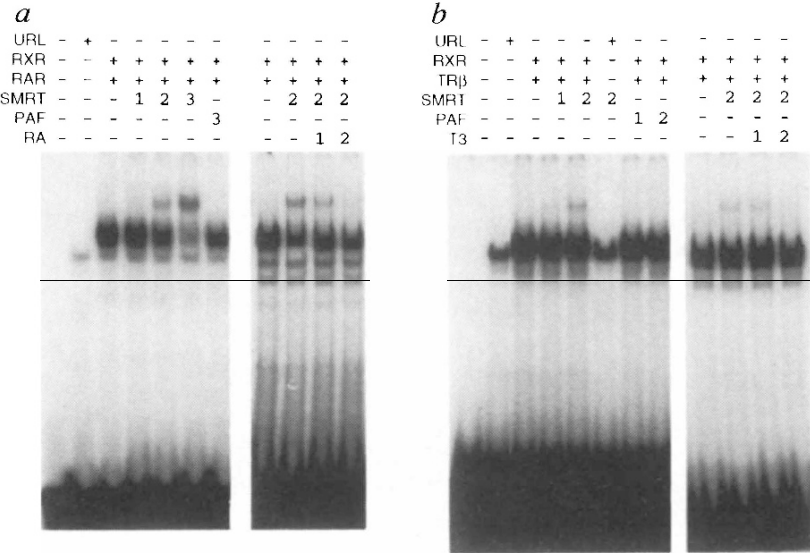
effects of these two receptors; the release of SMRT could be a prerequisite in ligand-dependent transactivation.

RAR and TR form heterodimers with RXR, resulting in a complex with high DNA-binding ability^{16, 18}. We tested whether SMRT could interact with receptor-DNA complexes. Figure 3a shows that SMRT forms a ternary complex with the RXR-RAR heterodimer on a RAR-response element and that addition of ligand releases SMRT from this complex. Similarly, SMRT forms a ternary complex with the RXR-TR heterodimer on a TR-response element and addition of T3 disrupts the formation of this complex (Fig. 3b). These data show that SMRT may be recruited to target promoters by interaction with DNA-binding unliganded receptors in a ligand-reversible manner.

It has been shown that overexpression of RAR or TR can reverse the transcriptional silencing effect of the GAL4-DBD fusion of TR (GAL-TR) or RAR (GAL-RAR)^{19, 20}, presumably by competition for a limiting amount of a co-repressor. A similar effect is shown in Fig. 4a and b where overexpression of v-ErbA or RAR403 reverses the silencing effect of GAL-RAR and GAL-TR on the basal activity of a luciferase reporter. Co-expression of the full-length or the C-terminal domain of SMRT (C-SMRT) titrates out the v-ErbA or RAR403 competitor activity and re-endsows GAL-RAR and GAL-TR with silencing activity. As a control, neither v-ErbA nor SMRT show any effect on the transactivation activity of GAL-VP16 (Fig. 4c) or on the basal activity of GAL-RXR (Fig. 4a). Furthermore, the full-length SMRT but not C-SMRT can further suppress two- to threefold the basal level activity of promoters containing either RAR to TR response elements (data not shown). Thus, we propose that SMRT can functionally replace the transcriptional co-repressor of RAR and TR *in vivo*.

FIG. 2 a, Interaction of wild-type hRARα and deletion mutant RAR403 with SMRT. ³⁵S-labelled receptors were used as probes to hybridize immobilized GST-SMRT in the presence (10 μM) or absence of all-trans retinoic acid. The total bacteria extract expressing GST-RXR was included as a control. When quantified by phosphorimager, RAR403 shows a 4-fold stronger interaction with SMRT than does wild-type RAR. b, Interaction of the oncogenic v-ErbA and rTRα R160 mutant (TR160) with GST-SMRT in a far-western assay. When quantified by phosphorimager, the v-ErbA shows an 18-fold stronger interaction with SMRT than hTRβ, and the TR160 mutant shows a 10-fold lower signal than the rTRα. c, Autoradiograph showing the ³⁵S-labelled receptors on SDS-PAGE. RAR403: hRARα amino acids (aa) 1-403; ΔΔR: hRARα aa 154-462; ΔΔR403: hRARα aa 157-403; ΔΔTR160: LBD of rTRα with Pro 160 to Arg mutation; ΔΔTRα: rTRα aa 122-410; TR160: full-length rTRα with Pro 160 to Arg mutation.

FIG. 3 Interaction of SMRT with receptor-DNA complex and the effect of ligands. a, Interaction of SMRT with RXR-RAR heterodimer on a DR5 element comprising AGGTCA direct repeat spaced by five nucleotides in a gel retardation assay. The amounts of GST fusion protein added in this experiment are (1), 6 ng; (2), 60 ng; and (3), 600 ng. PAF is a PML-associated protein isolated from another yeast two-hybrid screen (J.D.C. et al., unpublished result). b, Interaction of SMRT with RXR-TR heterocomplex on a MLV TRE element. The amount of GST fusion protein added in this experiment are (1), 10 ng; (2) 100 ng. METHODS. Gel retardation assay: *in vitro* translated receptor or unprogrammed reticulocyte lysate (URL) was incubated with 1 μg poly(dIdC) on ice for 15 min in a total of 20 μl containing 75 mM KCl, 7.5% glycerol, 20 mM HEPES (pH 7.5), 2 mM DTT and 0.1% NP-40, with or without ligand as indicated (1, 10 nM; 2, 100 nM). A ³²P-labelled double-stranded oligonucleotide probe was added into the binding reaction (10,000 c.p.m. per reaction), and the reaction was further incubated for 20 min at room temperature. The protein-DNA complex was separated on a 5% native polyacrylamide gel at 150 volts.



If SMRT is the mediator of TR and RAR silencing, then direct recruitment of SMRT to a heterologous promoter should result in repression of the basal activity, as demonstrated before for the Ssn6-Tup1 co-repressor^{21,22}. We tested this idea by fusing SMRT to the GAL4 DBD and analysed its effect on the basal activity of the GAL4-dependent reporter. Figure 4d shows that both the full-length SMRT and its N-terminal domain can silence the basal promoter activity to a similar degree to the GAL-TR fusion, whereas the C-terminal receptor-interacting domain shows no silencing activity. Thus the silencing and receptor association domains appear to be functionally distinct. These

data suggest that SMRT, when it is recruited to a promoter by direct DNA binding or through association with an unliganded receptor, functions as a potent transcriptional repressor.

In conclusion, we have identified a novel co-repressor which mediates the transcriptional silencing of RAR and TR. Transcriptional silencing has been shown to be important in development, cell differentiation and in the oncogenic activity of *v-erbA*¹⁻³. In fact, *v-erbA* mutants that harbour the corresponding TR point mutation at position 160 neither suppress basal transcription nor are capable of oncogenic transformation^{14,15}. The function of SMRT as a silencing mediator of RAR and TR is analogous to several co-repressors in other systems²¹⁻²⁵. Because GAL-SMRT functions as a potent repressor when bound to DNA, the function of unliganded receptors may be to bring SMRT with them to target promoters via protein-protein interaction. Thus, the repressor function of TR and RAR involves direct interaction with SMRT, possibly in a way that stabilizes or promotes their interaction with the transcription factor TFIIB²⁶⁻²⁸. We have shown that ligand causes the dissociation of SMRT from the receptor, triggering the activation process. This could be followed (or be coincident) with an induced conformational change in the C-terminal trans-activation domain (τc , also called AF-2), allowing association with co-activators²⁹. Thus, as we have previously suggested¹⁵, the ligand-dependent activation of TR would represent two separable processes, including relief of repression and net activation. The isolation of SMRT now provides a tool for dissecting the molecular basis of *trans*-repression by nuclear hormone receptors.

Note added in proof: In an accompanying paper, Hörlein *et al.*³¹ propose the name TRAC (for thyroid-, retinoic acid-receptor-associated co-repressor) for the new family of proteins. □

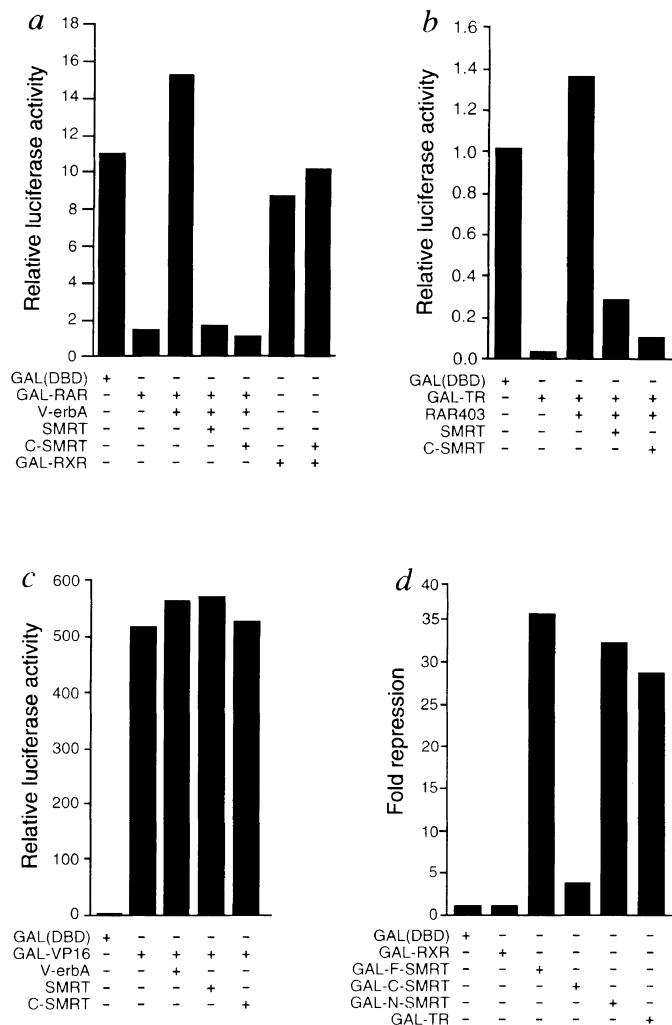


FIG. 4 Mediation of the silencing effect of hRAR α and hTR β by SMRT *in vivo*. **a**, Effect of v-ErbA and SMRT on the basal level promoter activity of GAL-RAR (GAL4 DBD-hRAR α 156–462) and GAL-RXR (GAL4 DBD fusion of hRXR α LBD). **b**, Effect of RAR403 truncation mutant and SMRT on the basal-level promoter activity of GAL-TR (GAL4 DBD-hTR β 173–456). **c**, v-ErbA and full-length SMRT or C-SMRT have no effect on GAL-VP16 activity. **d**, The intrinsic *trans*-repression domain of SMRT analysed by GAL4 DBD fusions. Fold repression was calculated by dividing the normalized luciferase activity transfected with the GAL4 DBD alone by those transfected with 30 ng GAL DBD fusion constructs. GAL-F-SMRT, GAL4 DBD fusion of full-length SMRT; GAL-N-SMRT, GAL4 DBD fusion of N terminus of SMRT (aa 1–981); GAL-C-SMRT, GAL4 DBD fusion of C-SMRT; GAL-TR, GAL4 DBD fusion of hTR β LBD. **METHODS.** Transient transfection assay. The CV-1 cells were plated in 24-well plates at a density of 50,000 cells per well. The expression plasmids were transfected into cells by lipofection. In each transfection, 5 ng GAL-RAR and 15 ng v-erbA or SMRT were used together with 150 ng reporter construct containing 4 copies of GAL4 binding sites in front of a minimal thymidine-kinase promoter and a CMX- β -gal construct as an internal control. The relative luciferase activity was calculated by normalizing to the β -gal activity.

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